Software Requirements Specification

for

Stipend Point Allocator

Prepared by-

Abhay Bamboriya 0801CS221002

Submitted to –

Ashwini Sharma Ma'am

Mamta Gupta Ma'am

Table of Contents

Introduction **3**
1.1 Purpose 3
1.2 Document Conventions 3
1.3 Scope 3
2. Overall Description **4**
2.1 Product Perspective 4
2.2 Product Functions 4
2.3 User Classes and Characteristics 5
3. Functional Requirements 6
3.1 User Requirements 6
4. External Interface Requirements **6**
4.1 User Interfaces 6
4.2 Hardware Interfaces 6
4.3 Software Interfaces 6
5. Use Cases **7**
5.1 Upload Pdf's 7
5.2 Download Results 9
6. Other Nonfunctional Requirements **10**
6.1 Performance 10
6.2 Usability 10
6.3 Security 10
7. Input Format **10**
8. Output Format **10**
9. Example Input and Output Format **10**

1. Introduction

1.1 Purpose

This document provides a comprehensive description of the "Stipend Point Allocator" web application, which aims to allocate points to stipends based on user-defined ranges. The application features file uploads for processing stipend data, OCR (Optical Character Recognition) for extracting text from images, and PDF generation for exporting reports. Additionally, the backend API facilitates file uploads and serves as the integration point for various functionalities.

1.2 Document Conventions

This document follows the IEEE standards for software requirements specifications. Key elements include:

- **Bold Text**: For section headings and important terms.
- *Italics*: For emphasizing specific details.
- Numbered Lists: For prioritized or sequential items.

1.3 Scope

The application supports stipend management, file uploads (CSV), dynamic stipend range allocation, OCR integration for document scanning, and report generation in PDF format. The front-end is built using HTML, JavaScript, and CSS, while the_backend_uses_Express.js.

1.4Refrences

- **Optical_character_recognition_Wikipedia**

2. Overall Description

2.1 Product Perspective

The web-based system integrates CSV file processing, stipend point allocation, document scanning, and PDF generation. The front end handles user interactions and file uploads, while the back end processes data and manages file storage.

2.2 Product Functions

The major functions of the E-learning platform include:

- Uploading CSV files containing stipend data.
- Processing stipend data for point allocation based on defined ranges.
- Using a camera to capture images and extract text using OCR.
- Generating PDF reports of processed stipend data.
- Facilitating file sharing via services like File.io.

2.3 User Classes and Characteristics

- **General Users**: Users responsible for uploading stipend data and allocating points.
- **Administrators**: Users managing the system configurations and overseeing stipend allocation policies.

3. Functional Requirements

3.1 User Requirements

1. **File Upload**

   - Users must be able to upload a CSV file containing stipend data.
   - The system must process the file and extract the stipend information.

2. **Point Allocation**

   - The system must validate that ranges do not overlap and that there are no gaps between them.
   - Users must define stipend ranges and corresponding points through a dynamic input interface.

3. **OCR Functionality**

- Users can capture images via the camera, with OCR converting text from images.
- Extracted text must be displayed in the interface for verification.

4. **PDF Generation**

    - Users can generate a downloadable PDF of stipend data with allocated points.
    - The PDF must include a table displaying stipend ranges, associated points, and any relevant comments.

3.2 Non Functional Requirement

- Performance Requirements
- Scalability
- User Growth

4. External Interface Requirements

4.1 User Interfaces

- **Upload Form :** Allows users to select and upload CSV files.
- **Stipend Allocation Form:** Allows users to define stipend ranges and points.
- **Result Page**: Displays stipend data and provides download options for the PDF.

4.2 Hardware Interfaces

- The system requires a camera interface to capture images for OCR processing.

4.3 Software Interfaces

The application will use:

- **Frontend Libraries**: jsPDF for PDF generation, Tesseract.js for OCR, and autoTable for table formatting.
- **Backend Libraries**: Express.js for server-side processing and Multer for file handling.

5. Use Cases

5.1 Upload Stipend CSV File

Actor: User
Precondition: User is on the upload page.

Main Flow:

- User selects a CSV file.
- User clicks "Upload."
- User defines stipend ranges and assigns points.
- User clicks "Apply."
- The system processes the CSV and displays stipend data.

5.2 Download Results

Actor: User
Precondition: Data extraction has been completed.

Main Flow:

- User clicks "Generate PDF."
- The system generates and downloads the PDF report.

6. Other Nonfunctional Requirements

6.1 Performance

- The system should handle multiple stipend ranges without significant lag.O
- OCR processing should be completed within a few seconds for typical document sizes.

6.2 Usability

- The interface should be intuitive and responsive across devices.

6.3 Security

- All uploaded files should be validated and sanitized.

7. Input Format

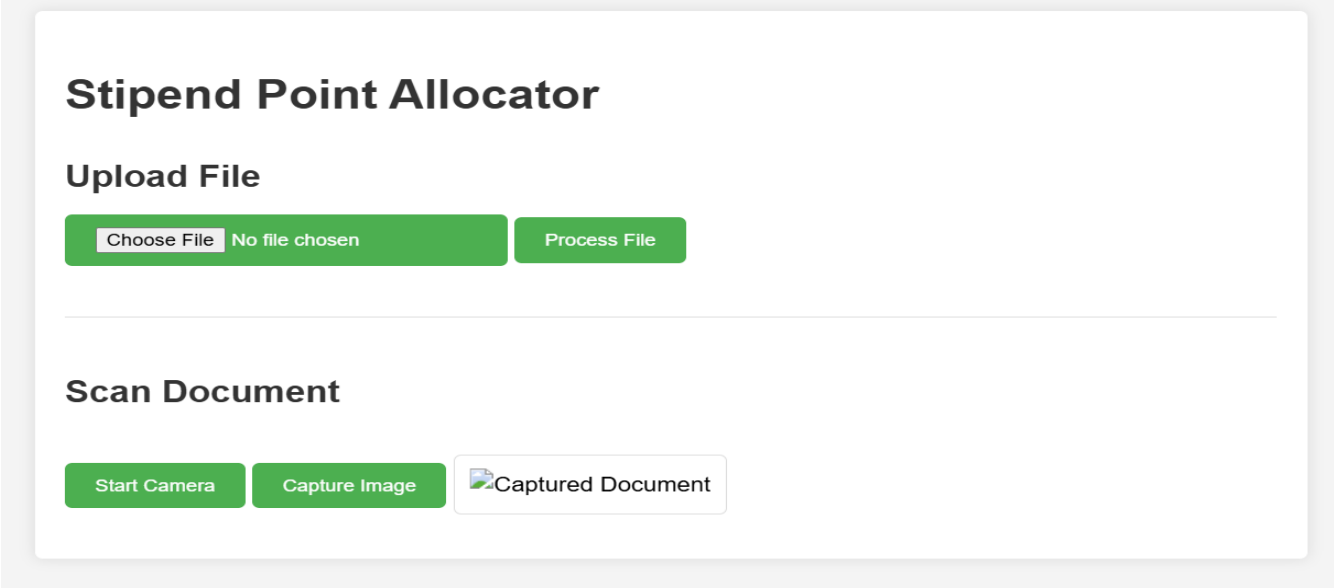
- **CSV File**: Contains stipend data in tabular form.
- **Camera Input**: Captures images to be processed by the OCR engine.
- **Range Values**: User-defined ranges for point allocation.

8. Output Format

- **PDF Report**: Contains a table of stipend data and allocated points, with columns for ranges, points, and any comments.

9. Example Input and Output Format

- **Example Input**



- **Example Output**

Range 1: 0 1200 - Points: 10 Remove

Range 2: 1200 2400 - Points: 20 Remove

Range 3: 2400 3600 - Points: 30 Remove

Range 4: 3600 4500 - Points: 40 Remove

Add Range | Apply Ranges | Download Updated PDF | Share on WhatsApp

Results

Name	Department	Position	Stipend	Start Date	Points
John Doe	Engineering	Intern	1200	2024-01-15	10
Jane Smith	Marketing	Assistant	2500	2023-11-01	30
Mike Johnson	Finance	Analyst	3800	2024-02-01	40
Sarah Brown	Human Resources	Coordinator	1800	2023-12-10	20
Chris Lee	IT	Technician	2200	2024-01-02	20
Emily Davis	Sales	Representative	3000	2023-10-20	30
Alex Wilson	Research	Associate	3500	2024-03-01	30
Olivia Taylor	Customer Service	Specialist	1500	2023-11-15	20
Daniel White	Operations	Manager	4500	2024-02-15	40
Sophie Martin	Design	Junior Designer	2000	2023-12-01	20